

annotated
version

MACHINE LEARNING COURSE · CS-433

Maximum Likelihood

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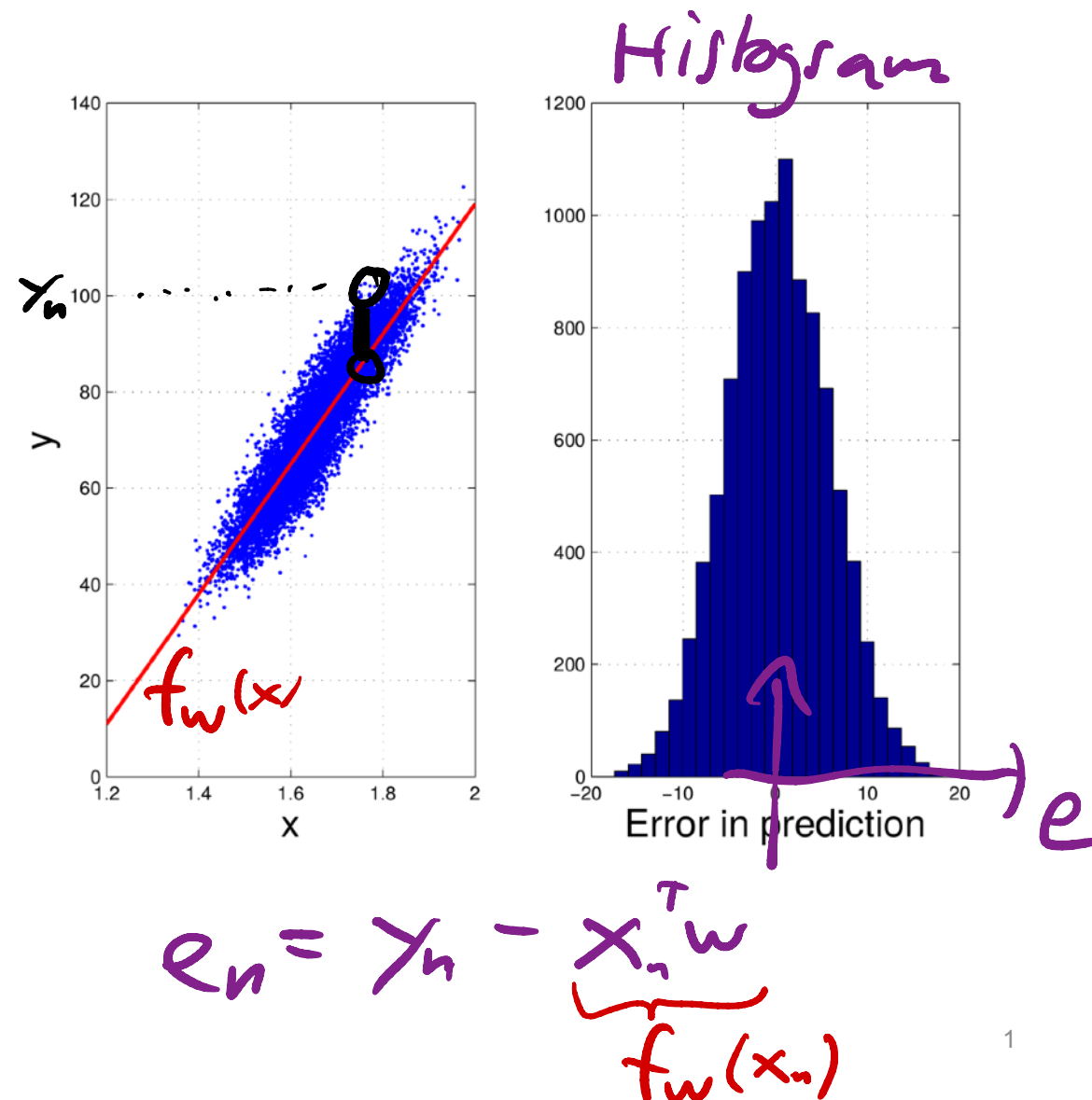
credits to Mohammad Emtiyaz Khan & Rüdiger Urbanke

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Motivation

In the previous lecture 3a we arrived at the least-squares problem in the following way: we postulated a particular cost function (square loss) and then, given data, found the model that minimizes this cost function.

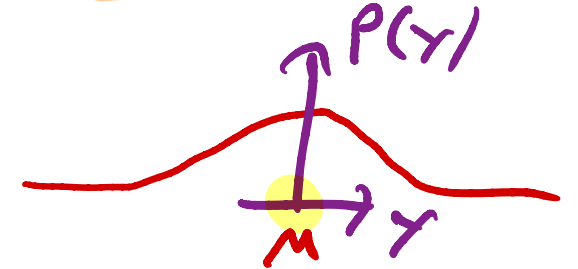
In the current lecture we will take an alternative route. The final answer will be the same, but our starting point will be probabilistic. In this way we find a second interpretation of the least-squares problem.



Gaussian distribution

Recall the definition of a Gaussian random **variable** in $\mathbb{R}$ with mean μ and variance σ^2 . It has a density of

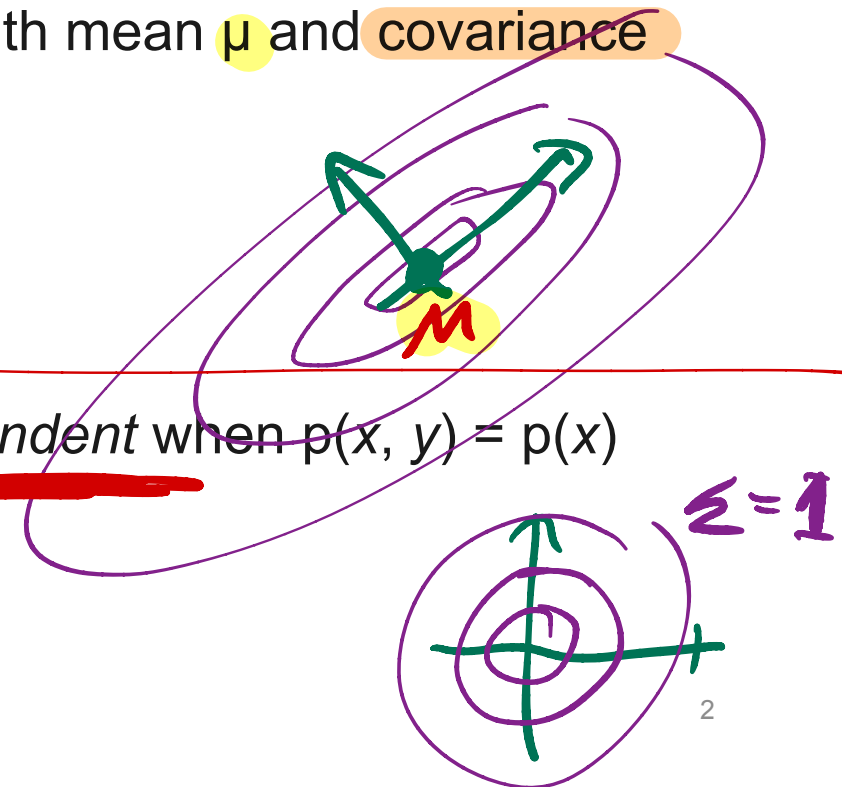
$$p(y \mid \mu, \sigma^2) = \mathcal{N}(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y - \mu)^2}{2\sigma^2}\right]$$



In a similar manner, the density of a Gaussian random **vector** with mean μ and covariance Σ (which must be a positive semi-definite matrix) is

$$\mathcal{N}(\mathbf{y} \mid \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^D \det(\Sigma)}} \exp\left[-\frac{1}{2}(\mathbf{y} - \mu)^\top \Sigma^{-1}(\mathbf{y} - \mu)\right]$$

Also recall that two random variables X and Y are called *independent* when $p(x, y) = p(x)p(y)$.



A probabilistic model for least-squares

We assume that our data is generated by the model,

$$y_n = \mathbf{x}_n^T \mathbf{w} + \epsilon_n$$

noise

ϵ_n gaussian

$$p(\epsilon_n) = \mathcal{N}(\epsilon_n | 0, \sigma^2)$$

$$\Leftrightarrow p(y_n) = \mathcal{N}(y_n | \mathbf{x}_n^T \mathbf{w}, \sigma^2)$$

where the ϵ_n (the noise) is a zero-mean Gaussian random variable with variance

σ^2 , and the noise added to the various samples is independent of each other, and independent of the input. Note that the model w is unknown.

The likelihood of the data

Given N samples, the **likelihood** of the data vector $y = (y_1, \dots, y_N)$ given the input X (each row is one input) and the model w is equal to

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = \prod_{n=1}^N \underbrace{p(y_n \mid \mathbf{x}_n, \mathbf{w})}_{\text{independence}} = \prod_{n=1}^N \mathcal{N}(y_n \mid \mathbf{x}_n^\top \mathbf{w}, \sigma^2)_{\text{assumption on noise}}$$

The probabilistic view point is that we should maximize this likelihood over the choice of model w . I.e., the “best” model is the one that maximizes this likelihood.

Defining cost with the log-likelihood

Instead of maximizing the likelihood, we can take the logarithm of the likelihood and maximize that instead. This expression is called the **log-likelihood** (LL).

$$\begin{aligned}\mathcal{L}_{\text{LL}}(\mathbf{w}) &:= \log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = -\frac{1}{2\sigma^2} \sum_{n=1}^N (y_n - \mathbf{x}_n^\top \mathbf{w})^2 + \text{cnst} \\ &= \log \prod_{n=1}^N p(y_n \mid x_n, w) \\ &= \log \prod_{n=1}^N \mathcal{N}(y_n \mid \mathbf{x}_n^\top \mathbf{w}, \sigma^2) \\ &= \log \prod_{n=1}^N \tilde{c} \exp\left(-\frac{(y_n - \mathbf{x}_n^\top \mathbf{w})^2}{2\sigma^2}\right) \\ &= \sum_{n=1}^N \log \exp(\dots) + \text{cnst} \\ &= \sum_{n=1}^N -\frac{1}{2\sigma^2} (y_n - \mathbf{x}_n^\top \mathbf{w})^2 + \text{cnst}\end{aligned}$$

Handwritten notes: "work" with an arrow pointing to the log-likelihood expression; "log p(y|x,w)" above the first step; "log" above the product steps.

Log-likelihood vs. MSE

Compare the LL to the MSE (mean squared error):

$$\max_w \mathcal{L}_{LL}(\mathbf{w}) = -\frac{1}{2\sigma^2} \sum_{n=1}^N (y_n - \mathbf{x}_n^\top \mathbf{w})^2 + \text{cnst}$$

indep of w

$$\min_w \mathcal{L}_{MSE}(\mathbf{w}) = \frac{1}{2N} \sum_{n=1}^N (y_n - \mathbf{x}_n^\top \mathbf{w})^2$$

$$\begin{aligned} & \text{||} \quad \arg\max_w \mathcal{L}_{LL} \\ & \arg\min_w \mathcal{L}_{MSE} \end{aligned}$$

Maximum-likelihood estimator (MLE)

It is clear that maximizing the LL is equivalent to minimizing the MSE:

$$\arg \min_{\mathbf{w}} \mathcal{L}_{\text{MSE}}(\mathbf{w}) = \arg \max_{\mathbf{w}} \mathcal{L}_{\text{LL}}(\mathbf{w})$$


This gives us another way to design cost functions.

MLE can also be interpreted as finding the model under which the observed data is most likely to have been generated from (probabilistically). This interpretation has some advantages that we discuss now.

Properties of MLE

$$\mathcal{L}_{LL} = \frac{1}{N} \sum_{n=1}^N \log p(y_n | x_n, w)$$

MLE is a *sample* approximation to the *expected log-likelihood*:

$$\mathcal{L}_{LL}(\mathbf{w}) \approx \mathbb{E}_{p(y, \mathbf{x})} [\log p(y | \mathbf{x}, \mathbf{w})]$$

max over w

w_{MLE}

$\propto$ dataset



w_{true}

MLE is **consistent**, i.e. it will give us the correct model assuming that we have a sufficient amount of data.

(can be proven under some weak conditions)

$$\underline{w_{MLE}} \xrightarrow{p} \underline{w_{true}} \text{ in probability}$$

Properties of MLE

Optional

The MLE is asymptotically normal, i.e.,

$$\underline{(\mathbf{w}_{\text{MLE}} - \mathbf{w}_{\text{true}})} \xrightarrow{d} \frac{1}{\sqrt{N}} \mathcal{N}(\mathbf{w}_{\text{MLE}} \mid 0, \mathbf{F}^{-1}(\mathbf{w}_{\text{true}}))$$

mean 0

where $\mathbf{F}(\mathbf{w}) = -\mathbb{E}_{p(\mathbf{y})} \left[\frac{\partial^2 \mathcal{L}}{\partial \mathbf{w} \partial \mathbf{w}^\top} \right]$

is the Fisher information.

MLE is **efficient**, i.e. it achieves the Cramer–Rao lower bound.

$$\text{Covariance}(\mathbf{w}_{\text{MLE}}) = \mathbf{F}^{-1}(\mathbf{w}_{\text{true}})$$

Another example

$$y_n \leftarrow x_n^T w + \varepsilon$$

We can replace the Gaussian distribution by a Laplace distribution.

$$\max p(y_n \mid \mathbf{x}_n, \mathbf{w}) = \frac{1}{2b} e^{-\frac{1}{b} |y_n - \mathbf{x}_n^T \mathbf{w}|}$$

Working through the same steps then leads to the MAE cost instead of the MSE.

$$\max_w \mathcal{L} \Leftrightarrow \min_w \sum_{n=1}^N |y_n - \mathbf{x}_n^T \mathbf{w}|$$

MAE

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